

JEE MAIN REVISION MASTER: SEQUENCES AND SERIES SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Sequences And Series - Core Application	32 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Sequences And Series - Core Application (Very High Yield — 32 PYQs)

Core Principle: Key problem pattern in Sequences And Series.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\alpha=3+4+8+9+13+14+\dots$ upto 40 terms. If $(\tan\beta)^{(\alpha/1020)}$ is a root of the equation $x^2+x-2=0, \beta \in (0, \pi/2)$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The sum of the first ten terms of an A.P. is 160 and the sum of the first two terms of a G.P. is 8. If the first term of the A.P. is equal ..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "Let the sum of the first n terms of an A.P. be $3n^2+5n$. Then the sum of squares of the first 10 terms of the A.P. is:..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

The sum of the first ten terms of an A.P. is 160 and the sum of the first two terms of a G.P. is 8. If the first term of the A.P. is equal to the common ratio of the G.P. and the first term of the G.P. is equal to common difference of the A.P., then the sum of all possible values of the first term of the G.P. is:

- (A) $(34 / 9)$ **[Correct]**
- (B) $(34 / 13)$
- (C) $(32 / 9)$
- (D) $(32 / 13)$

Correct Answer: A

Solution Analysis: Let the first term of the A.P. be a and common difference be d . Then, from the question: first term of the G.P. $= d$ common ratio of the G.P. $= a$. Now use the given conditions one by one. 1. Sum of first 10 terms of the A.P. We know, $S_{10} = \frac{10}{2}[2a + (10-1)d]$ So, $160 = 5(2a + 9d)$ $2a + 9d = 32$...(1) 2. Sum of first 2 terms of the G.P. The G.P. has first term d and common ratio a . So its first two terms are: d, da . Their sum is 8 : $d + da = 8$ $d(1+a) = 8$...(2) 3. Express d from equation (1) From $2a + 9d = 32$ we get $9d = 32 - 2a$ $d = (32 - 2a) / 9$ Substitute this in equation (2): $\frac{(32 - 2a)}{9}(1 + a) = 8$